

Graph Analysis of DiT Token Activations

Measured E-2--E-7 results

50 trajectories | 450 snapshots | 256 nodes | 1152 features

Main k=16; detailed k=2,4,8,16,32,64; large components ≥ 32 nodes

Fixed example: golden retriever / r00 / call25 / block14

Call	Block	Mean degree	Mean clustering	Mean lambda2
10	4	23.865	0.303	0.3726
10	14	23.921	0.303	0.3659
10	24	23.505	0.353	0.1175
25	4	23.910	0.301	0.3821
25	14	23.049	0.359	0.0975
25	24	23.377	0.477	0.0413
40	4	23.913	0.332	0.2208
40	14	22.104	0.467	0.0287
40	24	24.852	0.503	0.0525

315/450 connected at k=2; all connected by k=15.

17/450 main graphs lack a qualifying separated simple eigenvalue.

51/450 normalized Fiedler curves contain a decrease.

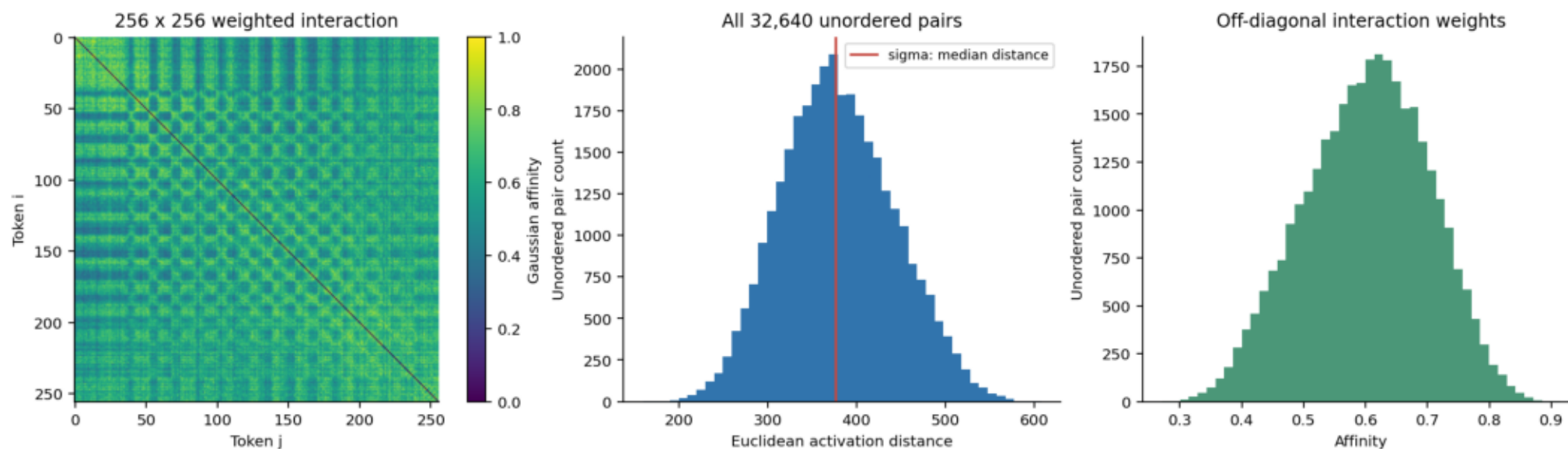
Results describe activation graphs and recorded updates.

They do not establish semantic segmentation, correction accuracy or speedup.

Exact method, source hashes and per-component data accompany this report.

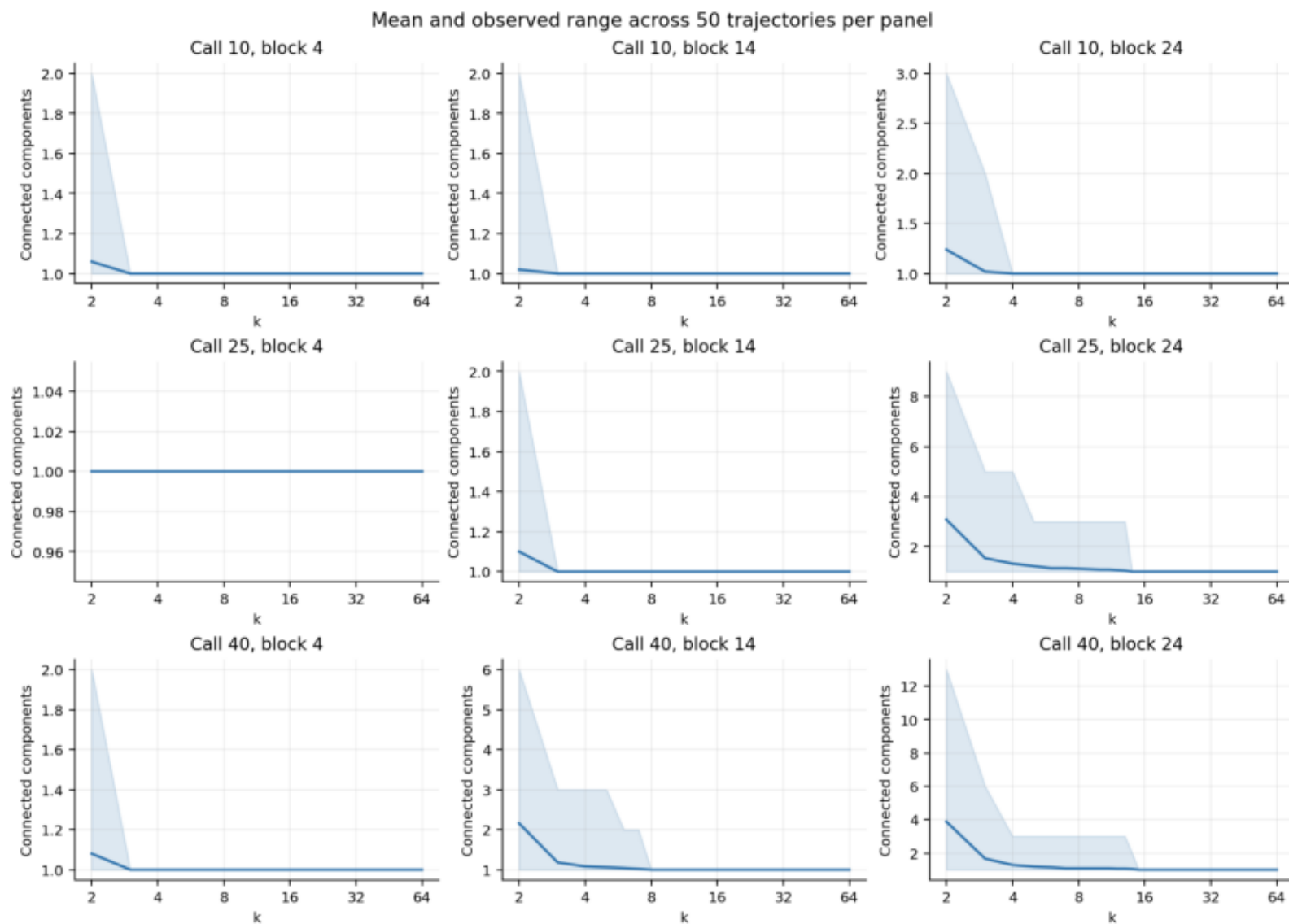
E-2: weighted pairwise interactions

All 1152 activation features; $S_{ij} = \exp(-||h_i - h_j||^2 / (2\sigma^2))$, diagonal zero. Sigma is the per-snapshot median Euclidean distance, fixed over k. 450 symmetric 256 x 256 matrices; no PCA, updates or spatial attributes enter graph construction.



E-3: kNN connectivity

Union of directed kNN edges, original Gaussian edge weights retained. Scan all integers $k=2..64$. 315/450 connected at $k=2$; all 450 connected by $k=15$. Shading is observed range, not a confidence interval. k_c is the smallest connected value in the scanned range.

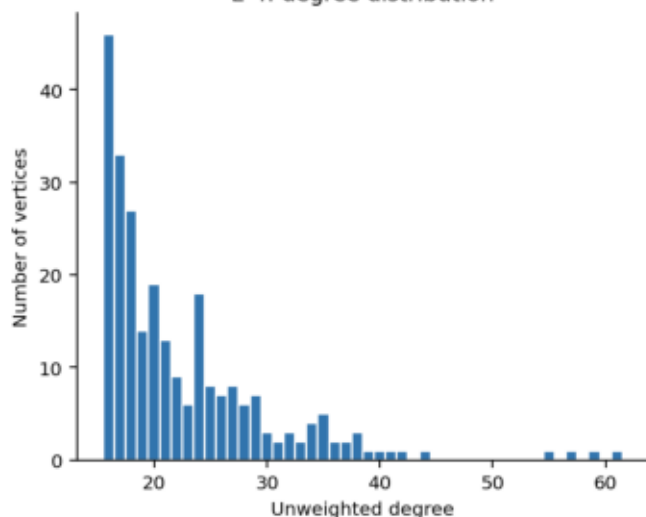


E-4 / E-5: representative graph

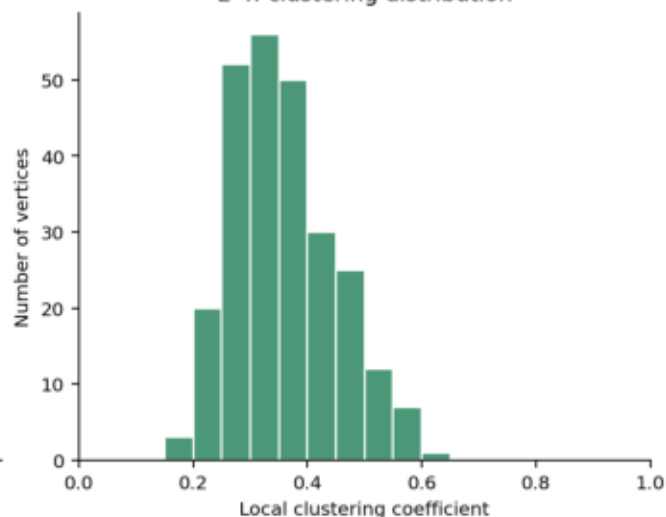
k=16: mean degree 22.508, mean clustering 0.359, lambda2=0.071018; selected q3=0.225608, 2 strict nodal domains. Degree is right-skewed; clustering mildly right-skewed. All detailed large components have saved plots.

imagenet-0207-r00 | call 25 | block 14 | k=16 | component 0 | n=256

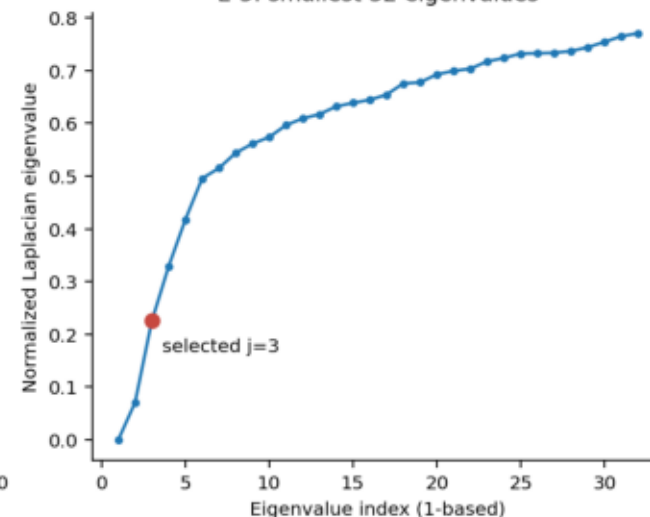
E-4: degree distribution



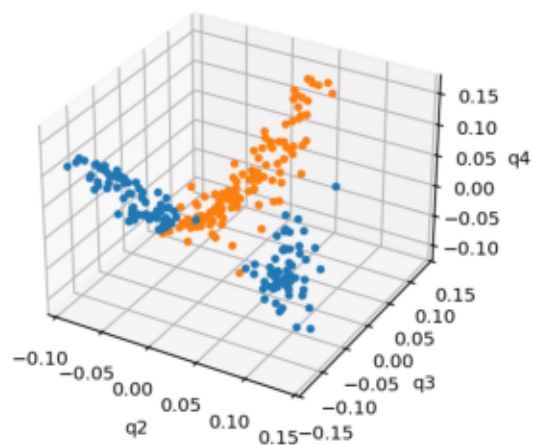
E-4: clustering distribution



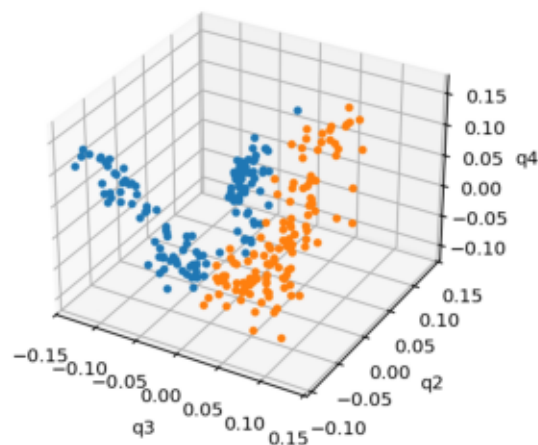
E-5: smallest 32 eigenvalues



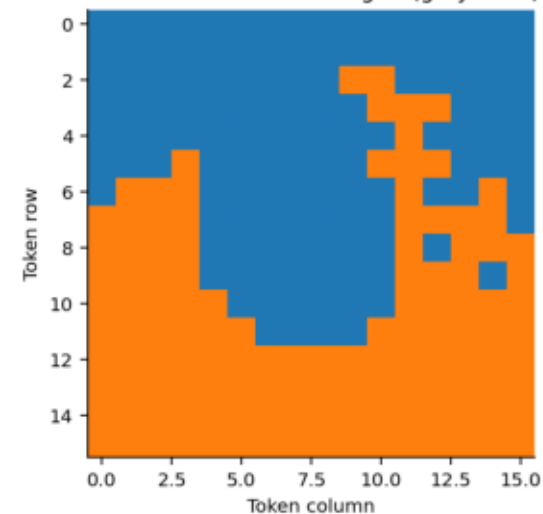
E-5: invariant subspace (q2,q3,q4)



E-5: 2 strict nodal domains

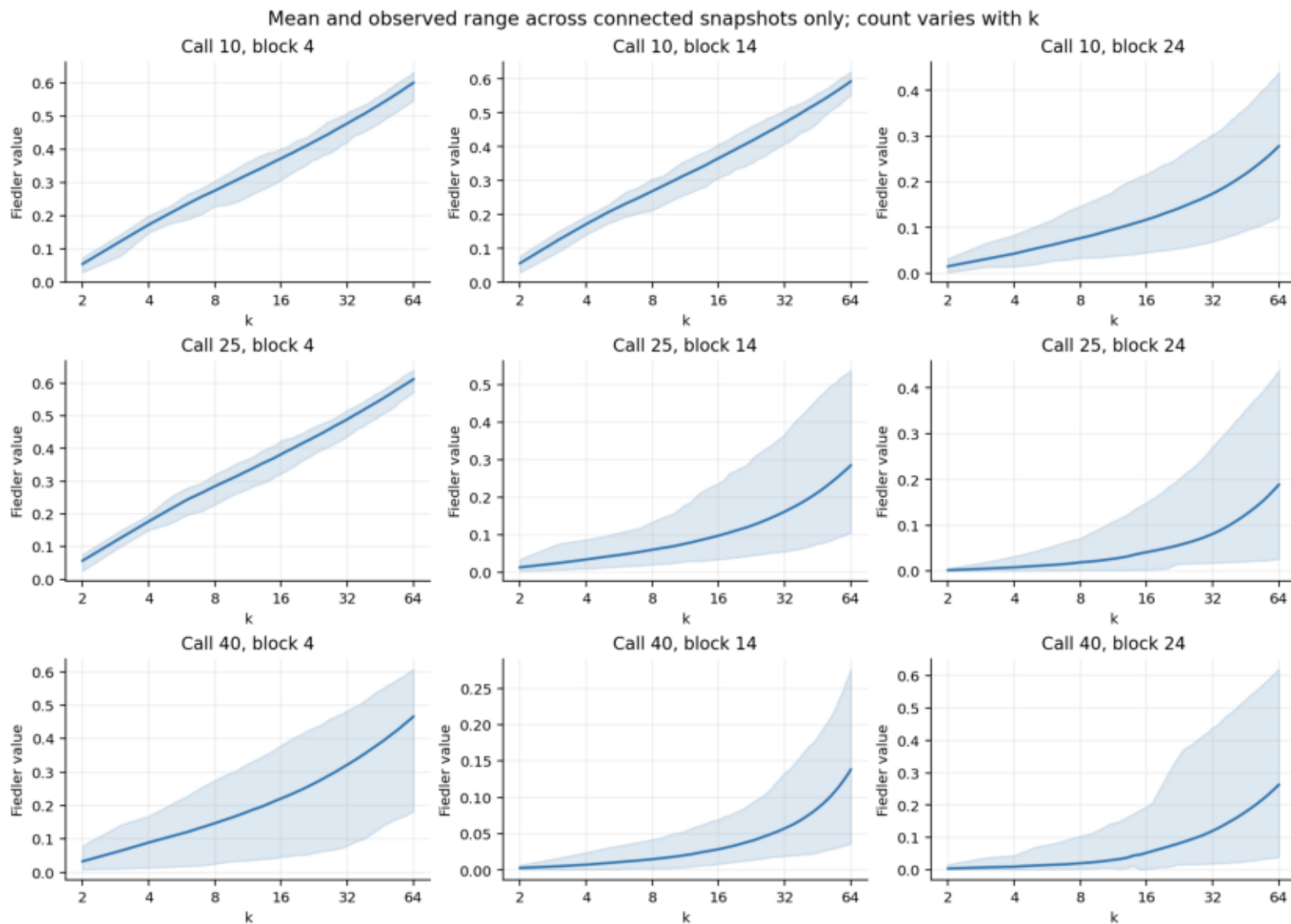


Nodal domains on token grid (gray: zero)



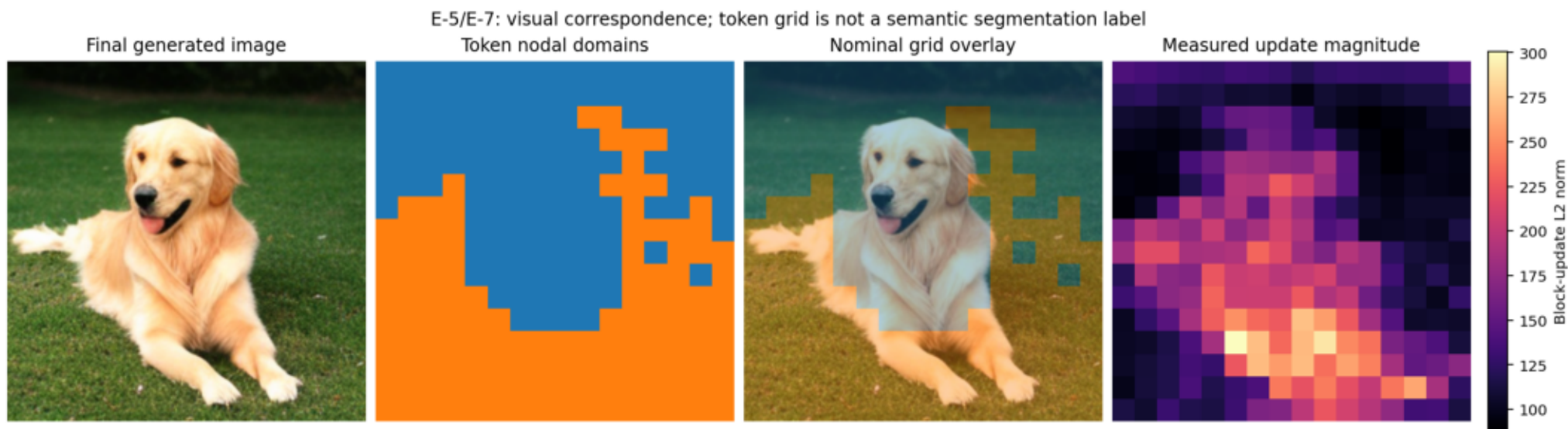
E-5: normalized-Laplacian Fiedler curves

$L_{\text{sym}} = I - D^{-1/2} W D^{-1/2}$. Plot connected snapshots only; their count can vary with k . 51/450 curves contain at least one decrease. 17/450 main graphs have no eigenvalue satisfying the predetermined two-sided separation rule; no nodal partition is fabricated for those cases.



E-5 / E-7: correspondence with original data

Two domains contain 131 and 125 tokens; mean update norms 143.73 and 159.76. Within-domain variation remains large; grouping explains only 2.53% of update-norm variance. The nominal token-grid overlay does not follow the dog's semantic boundary accurately.



E-6: optional two-cluster k-means baseline

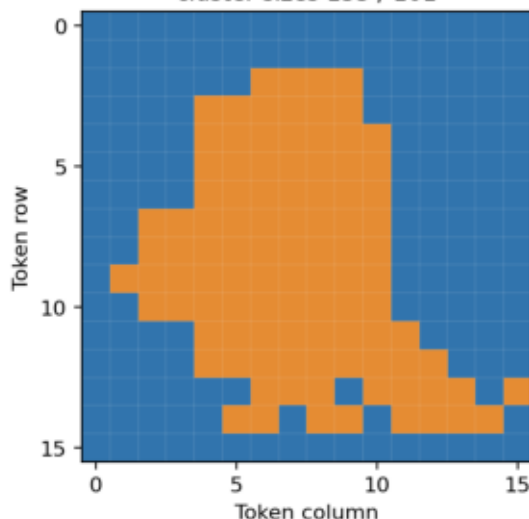
Original 1152 features; five fixed-seed initializations. Normalized cut: k-means 0.2047, nodal 0.3622. Partition ARI=0.0119; this is partition agreement, not semantic accuracy.

E-6: imagenet-0207-r00 | call 25 | block 14 | k=16

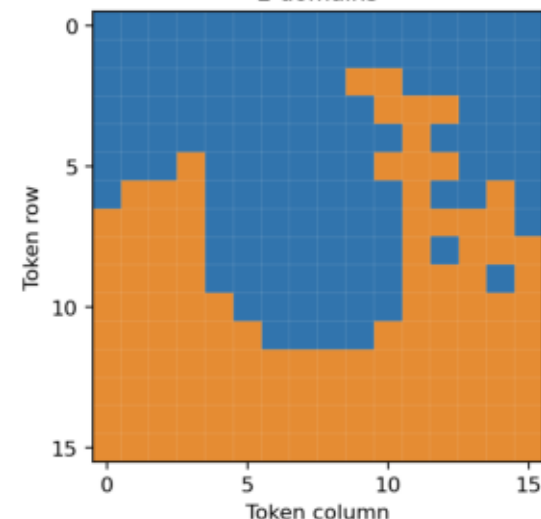
Generated image: golden retriever



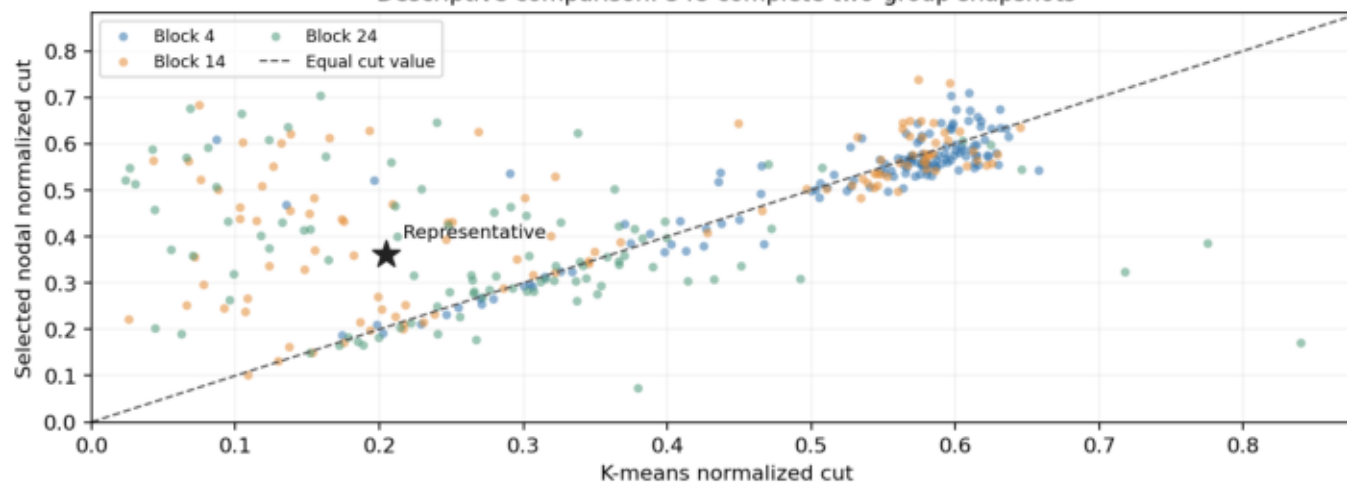
K-means, K=2
cluster sizes 155 / 101



Selected nodal domains, q3
2 domains



Descriptive comparison: 348 complete two-group snapshots



Representative comparison

K-means Ncut: 0.2047
Nodal Ncut: 0.3622
Partition ARI: 0.0119

K-means: 5 fixed starts,
50 iterations, 1152 channels.

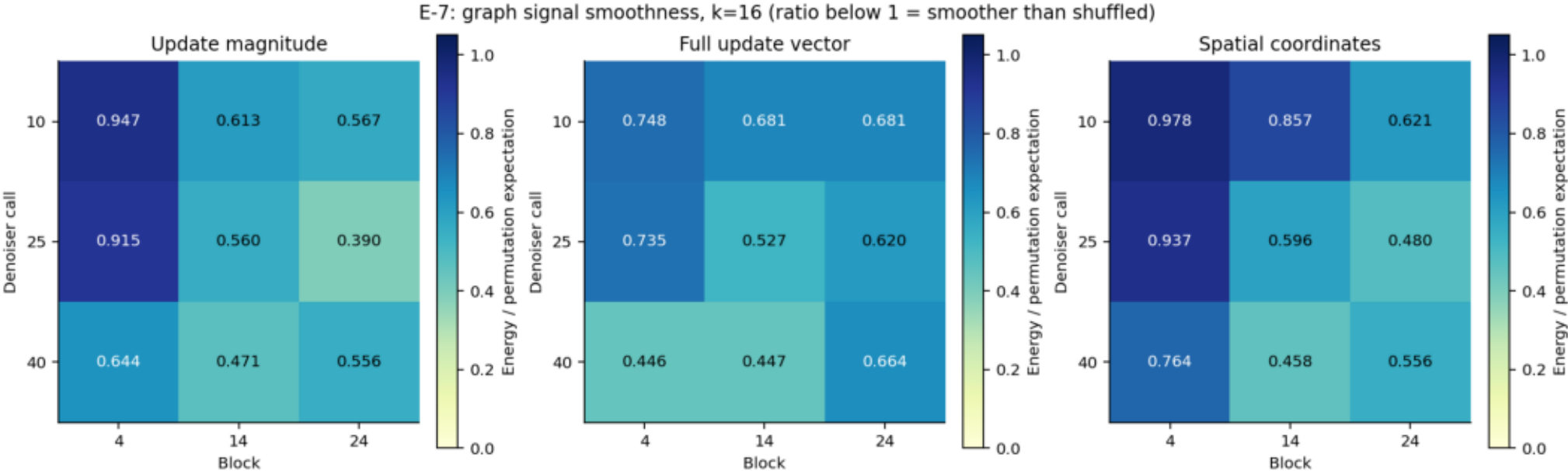
Nodal vector selected by eigenvalue gaps;
its partition does not optimize Ncut.

Blue: group 0; orange: group 1.
Labels are local; no semantic ground truth.

Scatter points share 50 trajectories;
they are descriptive observations.

E-7: graph signals and measured updates

Mean energy/null ratios at k=16: update magnitude 0.629, vector update 0.617, spatial coordinates 0.694. Edge-weighted update cosine 0.676 vs all-pairs 0.510. Ratios below 1 indicate stronger graph smoothness than random node assignment. Differences between conditions remain substantial.



E-7: preliminary cross-layer structure

Top-16 neighbor retention: 4->14=0.505; 14->24=0.539; 4->24=0.288. Uniform independent-set reference=16/255=0.063. This is descriptive structural evidence, potentially partly due to spatial coherence; cross-layer correction performance remains unmeasured.

